Rotor speeds from ego-noise: the wrap-up

Figures and tables on one frozen protocol

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2026-08-25

The question and the paper

Can we get per-rotor speeds from onboard audio alone?

- Per-rotor speeds from onboard audio: each rotor at rate f leaves harmonics at $f, 2f, 3f, \ldots$
- No published direct method exists — we build the baseline suite ourselves.
- A 26-variant architecture search gives three winner architectures.
- Augmentations and a full-envelope real training regime repair what models actually read.
- A blind two-stage classical tracker sets the cruise precision ceiling, at high CPU cost.
- Synthetic data supplies coverage, not realism — and loses its value once real data supplies the same coverage.

Data split

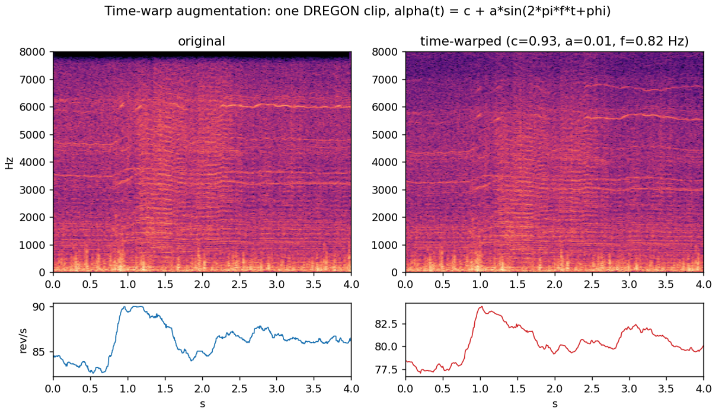
Split	DREGON (small quad)	MD2 (DJI M100)	Speech
Training	free-flight and hover recordings, refined per-rotor labels	FLY124, FLY125 (recalibrated telemetry)	LibriSpeech
Validation (frozen)	8 held-out slices, full envelope (ground, ramps, cruise)	29 held-out slices from FLY124/ FLY125	—
Test (reserved, untouched)	free-flight speech-high room1, whitenoise-high room1	FLY103, FLY108 (calibrated, frames published)	—

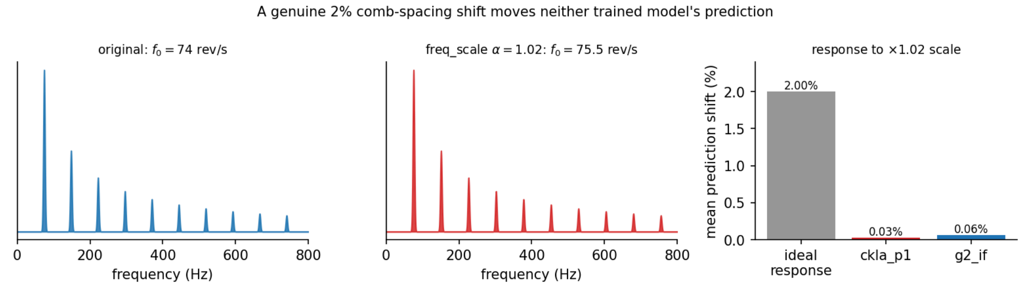
The validation set: 37 clips $\times$ 8 channels $\times$ 8 s. Per-frame Hungarian matching of 4 predicted to 4 true speeds; errors pooled per regime — **zero** (all rotors stopped), **ramps**, **flight** (all rotors $\geq$ 45 rev/s) — and overall (frame-weighted).

Training regimes

Regime	Full description
R1 — architecture search	Online mixing: random 1-s noise chunks from the real pool, random speech overlay at random SNR, new mixtures every epoch. Augmentations after a 50k-sample warm-up: random gain, polarity flip, channel drop (each $p = 0.5$). No silence, no label-transforming augmentations.
R2 — full-envelope real (final real-only)	R1 mixing plus: **silence arm** — 16.7% of chunks are synthetic room tone / colored noise / low-frequency rumble with all-zero speed labels; **SNR reference floor** — quiet chunks are not amplified to the target SNR (reference RMS floor 0.02), so level stops predicting speed; label-transforming **frequency-scale** and **time-warp** augmentations (labels scaled/warped together with audio).
R3 — generated + comb curriculum	Stage 1: pre-train on synthetic noise from the neural generator plus the analytic comb, full-envelope speed trajectories with exact silence. Stage 2: fine-tune under R2. Same optimizer, loss, budget.
R4 — comb-only curriculum	R3 with the neural generator removed: stage 1 is the analytic comb alone, stage 2 is R2.
R5 — mixed, one stage	One pool, one stage: real 1/2, generated 1/4, comb 1/4, plus the R2 silence arm; the same augmentation schedule. No curriculum.

Label-transforming augmentations



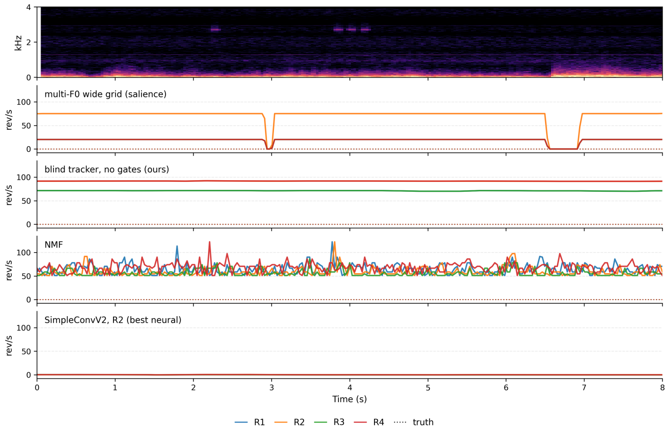


Time-warp: resample audio and speed labels on one warped clock ($\alpha \leq 1.12$) — new speed trajectories from the same recording.

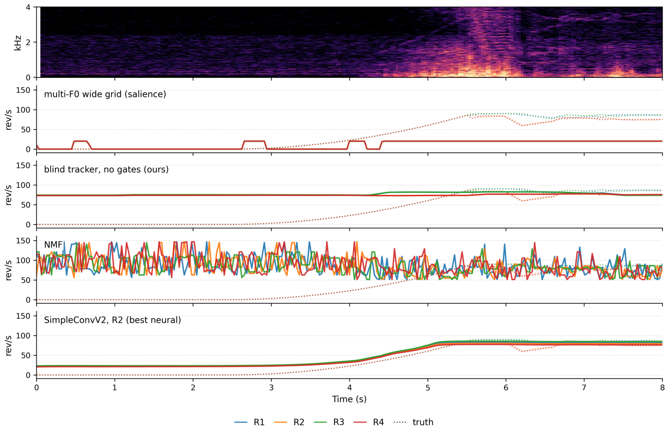
Frequency-scale: scale the spectrum and the labels by the same α — a genuine comb-spacing change the model must follow.

The three regimes of the validation set

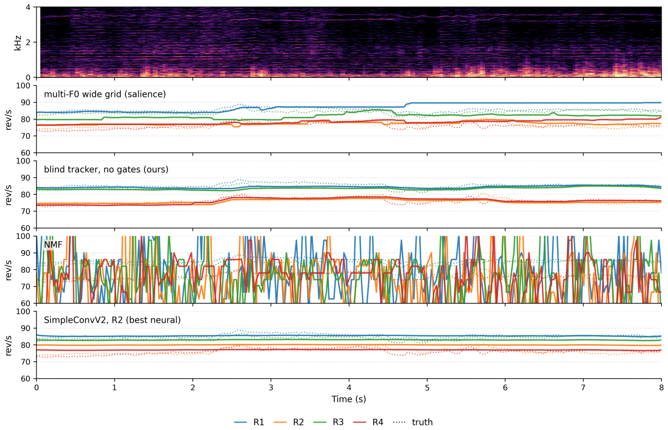
zero — stopped rotors



ramps — start/stop transition



flight — cruise



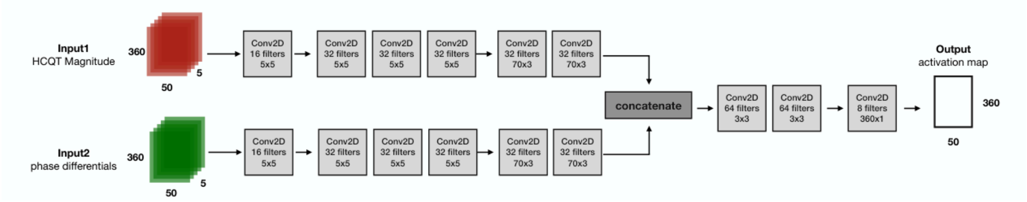
Validation slices: spectrogram on top, per-rotor speed tracks below. The same three clips return full-size later in the deck.

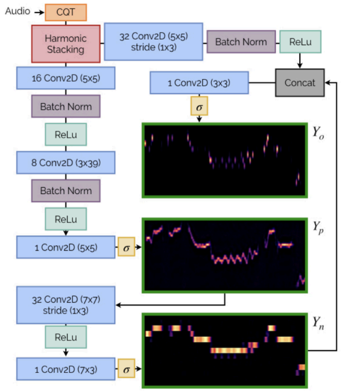
Classical pitch-tracking baselines

Method	What it is	zero	ramps	cruise	all
PYIN ×4	probabilistic YIN, greedy 4-peak pick	90.8	46.7	34.1	43.0
Cepstral	cepstral peak picking, greedy	95.5	67.5	19.6	35.7
HPS	harmonic product spectrum, greedy	64.1	27.6	20.9	27.3
Matched filter	comb-template bank correlation	87.2	66.8	30.4	42.5
NMF	non-negative factorization, harmonic dictionary	83.8	59.5	**8.1**	24.6
IHC	inverse harmonic clustering (Björkman & Elvander)	68.8	28.3	16.3	24.6

PIT MAE, rev/s. NMF is the best training-free cruise tracker (8.1). Every method floors at its grid boundary on stopped rotors — silence is unrepresentable for a pitch grid that starts at 50 rev/s.

Salience baselines: what they are

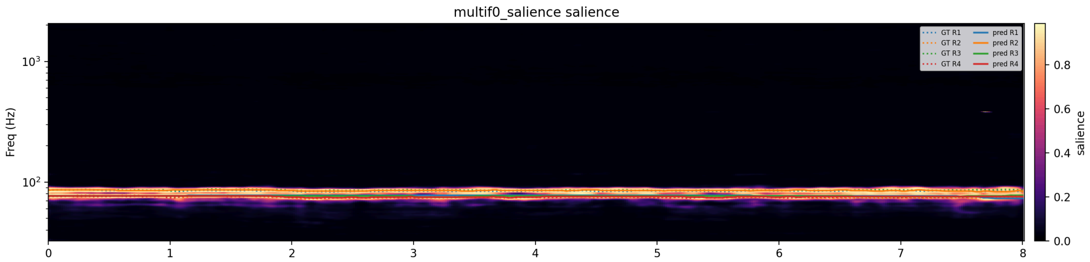


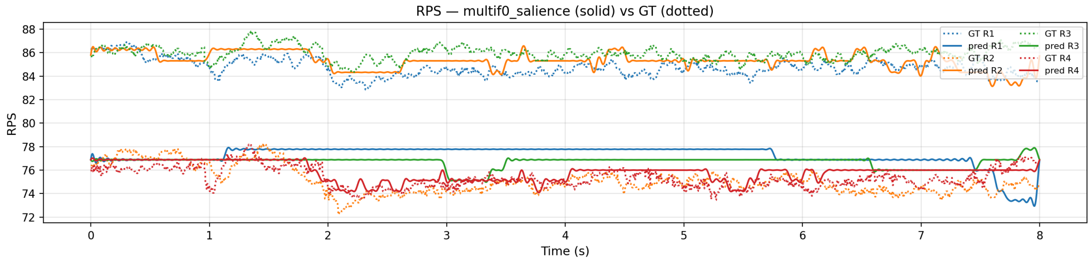


Multi-F0 CNN (Cuesta et al.): HCQT input, per-bin salience map, trained with BCE on the comb positions; Hungarian tracking decodes 4 speed tracks from the map.

Basic Pitch (Spotify): lightweight polyphonic pitch model, ported to PyTorch; same salience-map decoding. Both retrained under the same R2 stream and augmentations as our models.

Salience baselines: outputs and results



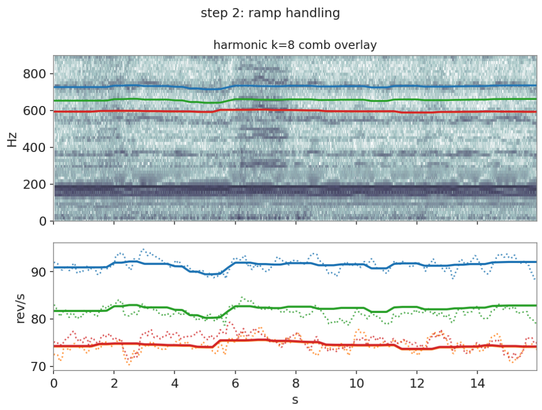


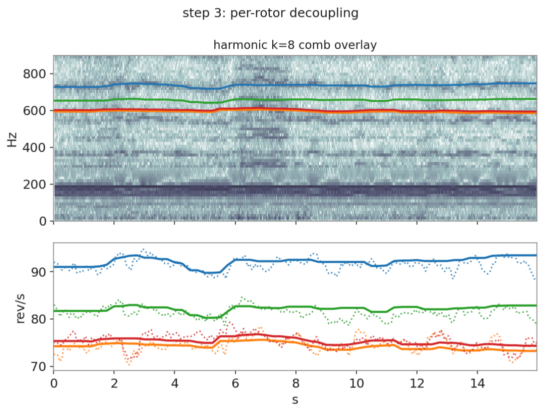
Multi-F0 salience map (top) and decoded tracks (bottom) on a cruise clip.

Retrained model	zero	ramps	cruise	all
Basic Pitch	34.0	13.3	31.7	29.5
multi-F0, standard grid	52.8	21.0	4.0	12.5
multi-F0, wide fine grid	48.2	16.1	4.7	11.7

Cruise-capable (4.0 rev/s) and silence-blind: the salience map lights up on room rumble, and no retraining regime fixes it — a model-class limitation, not a data one.

Our classical tracking: blind two-stage annotation



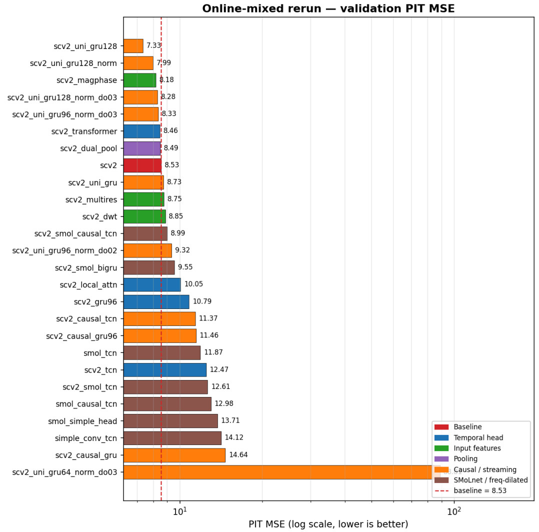


Stage 1 — Viterbi ridge-seeking: comb-score lattice over candidate speeds; joint 8-channel dynamic programming picks the smoothest high-score path per rotor.

Stage 2 — phase-increment Kalman refinement: demodulate each harmonic at the current estimate, read the phase increments, fuse them with SNR-dependent weights, smooth, repeat coarse-to-fine.

Zero gate: a window is accepted only when the comb score clears its off-comb floor (margin and clearance gates); a refused window emits zero speeds. Gate-free: cruise MAE **2.27** rev/s (best of all methods). Gated: zero MAE **0.01**, at the price of refusing hard flight windows.

Architecture search: 26 variants, 52 runs



The R1 sweep leaderboard (online-mixing arm): six variant groups over one conv trunk — temporal heads, input features, pooling, causal heads, dilated backbones.

STFT front-end (1 s chunk)
magnitude / +IF / synchrosqueezed

↓

residual conv encoder
squeeze-and-excitation blocks

↓

learned frequency-attention pool

↓

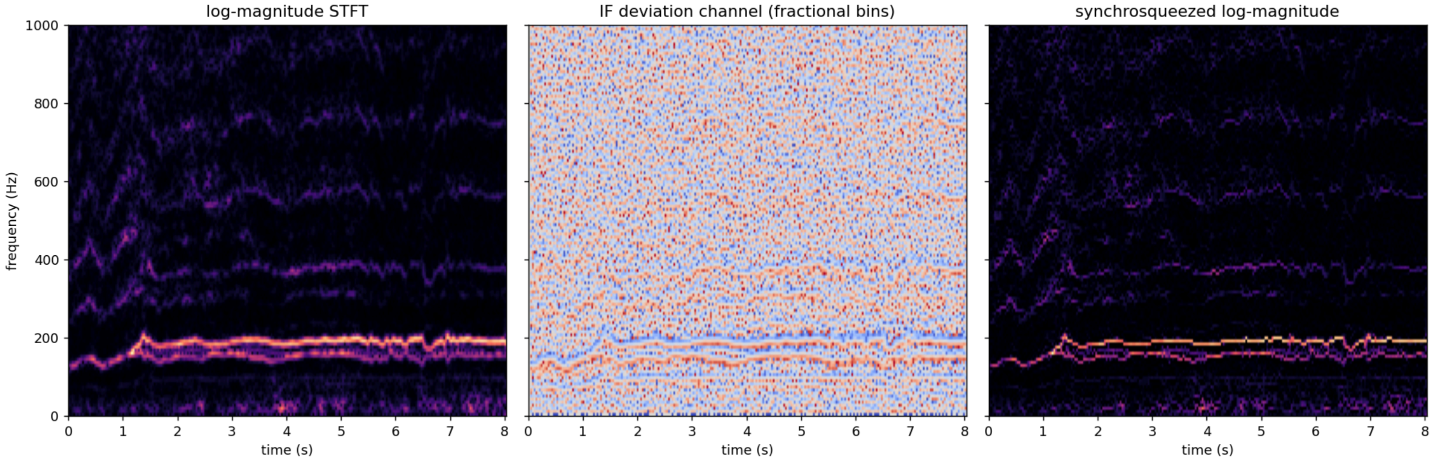
BiGRU SimpleConvV2	**Transformer** head	**causal GRU-128** streaming

↓

4 speeds per frame, PIT-MSE

The shared trunk and the three winner heads kept for the paper.

Spectral front-ends



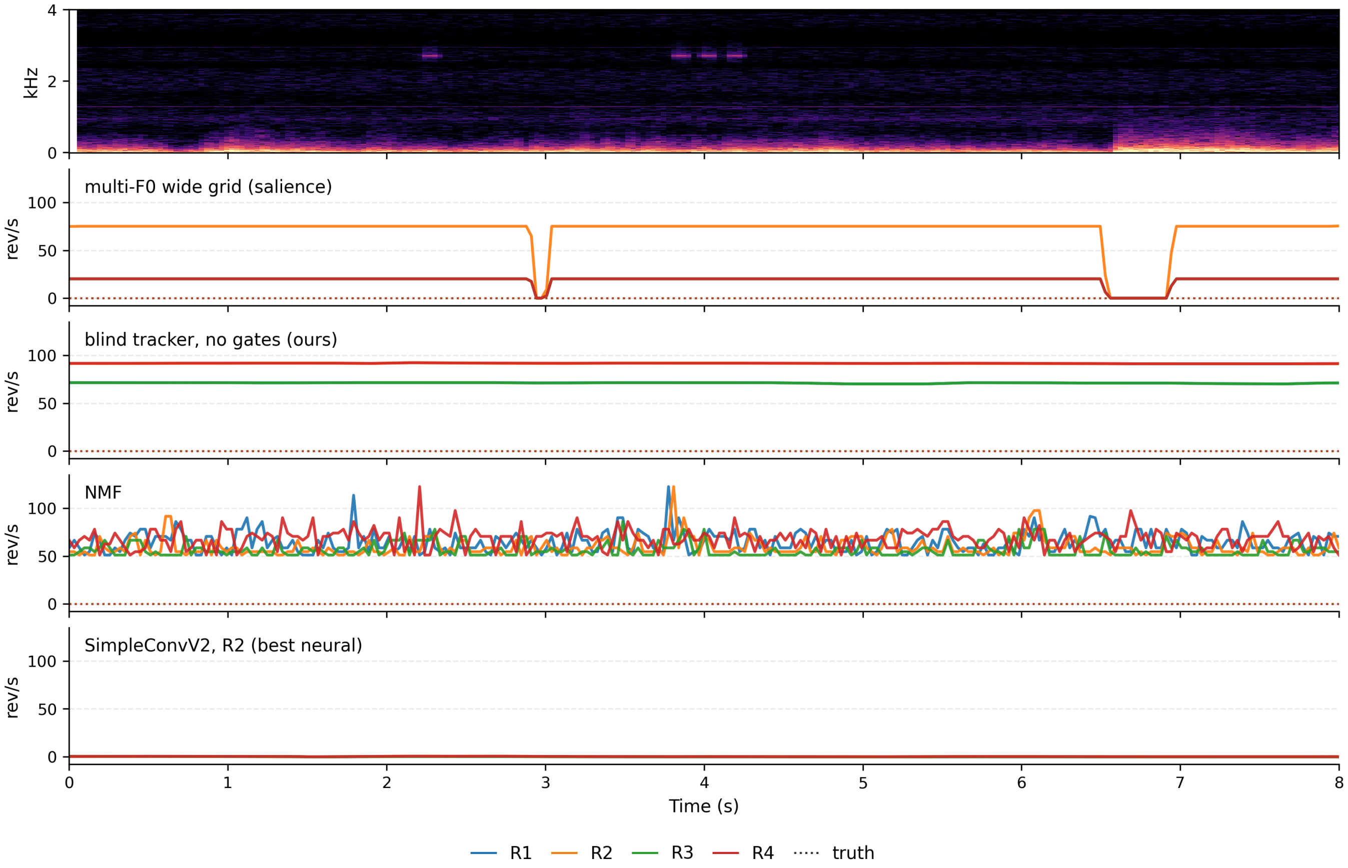
One cruise validation clip through the three front-ends. **Log-magnitude**: the default; comb teeth blurred by window leakage. **IF deviation**: per-bin instantaneous-frequency offset in fractional bins phase information the magnitude discards; zero-mean on noise, structured on tones. **Synchrosqueezed magnitude**: bin power reassigned to the frequency where its phase actually advances — leakage collapses back into sharp ridges, one channel. Best per architecture: IF for SimpleConvV2, magnitude for the Transformer, synchrosqueezed for the causal GRU.

The leaderboard: every family on one protocol

Method	zero	ramps	cruise	all	cost /audio-s
NMF (best training-free)	83.8	59.5	8.1	24.6	0.1 CPU-s
multi-F0 wide grid (best salience)	48.2	16.1	4.7	11.7	2 CPU-s
blind tracker, no gates	79.4	39.1	**2.27**	17.0	9.9 CPU-s
blind tracker, gates, refusal $\to$ 0	**0.01**	29.8	48.4	39.7	9.9 CPU-s
SimpleConvV2 (BiGRU), R2	**3.4**	**4.2**	2.35	**2.72**	0.25 CPU-s
Transformer head, R2	5.5	5.1	2.65	3.35	0.25 CPU-s
causal GRU-128, R2	6.0	8.2	3.06	4.12	0.25 CPU-s

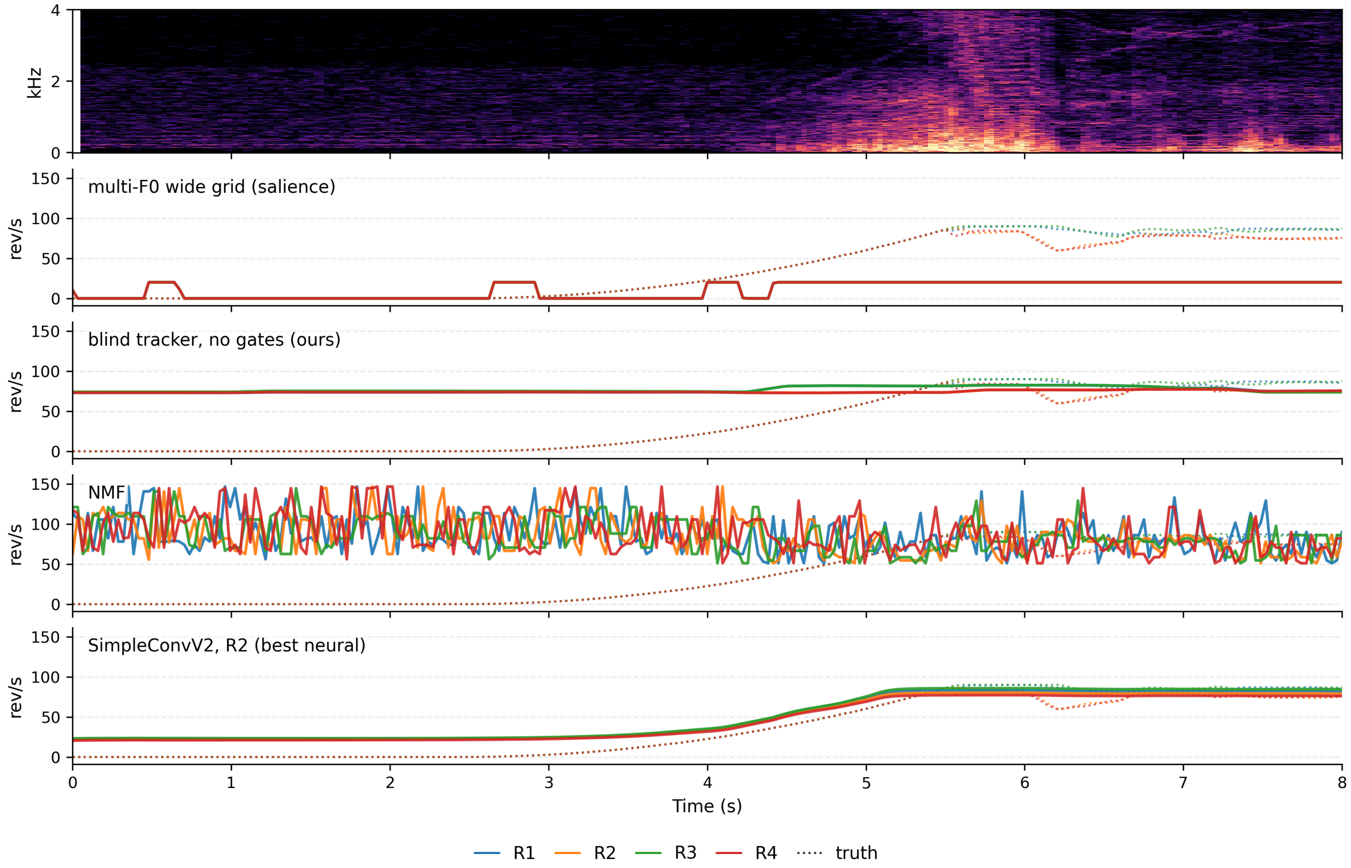
PIT MAE, rev/s. The regression models win every column except cruise, where the gate-free blind tracker stays ahead (2.27 vs 2.35) at $\sim$40$\times$ the cost. Only the gated tracker and the R2 models decide silence correctly.

Outputs: stopped rotors



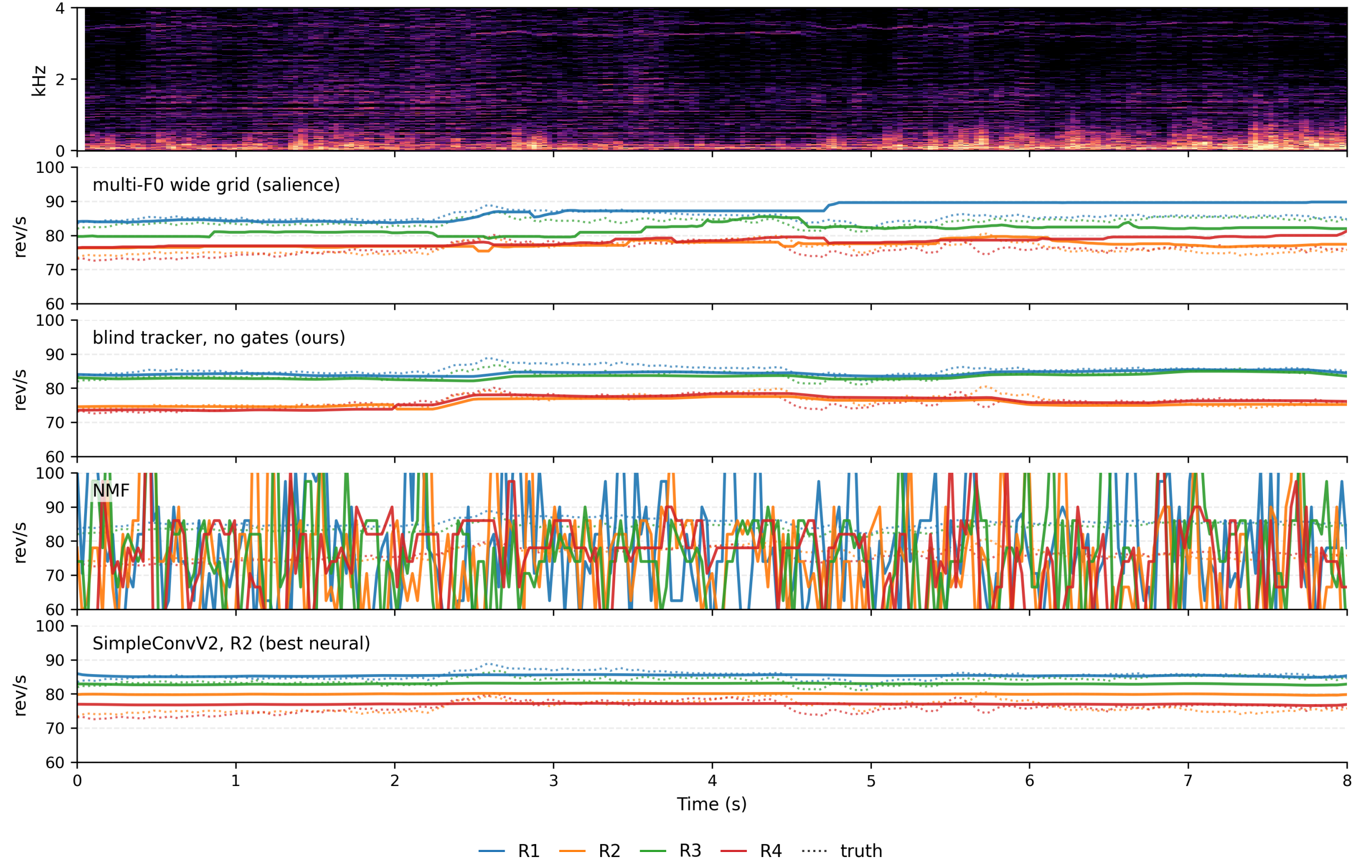
Rumble-heavy silence clip. The R2 model holds zero. NMF floors at its grid boundary, the salience model lights up on the rumble, and the gate-free tracker locks onto false ridges (the gates decode this window to zero).

Outputs: start/stop transition



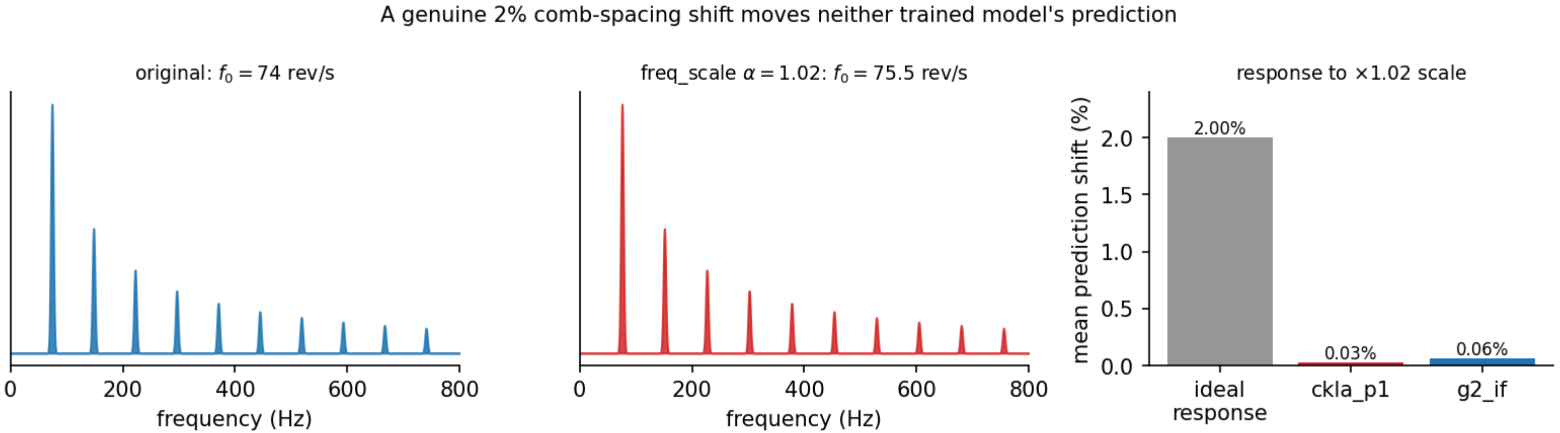
Ramp clip. The R2 model follows the ramp through the comb collapse. The salience model stays near its floor, NMF answers noise below its grid, and the blind tracker holds a false cruise lock until the true comb appears.

Outputs: cruise



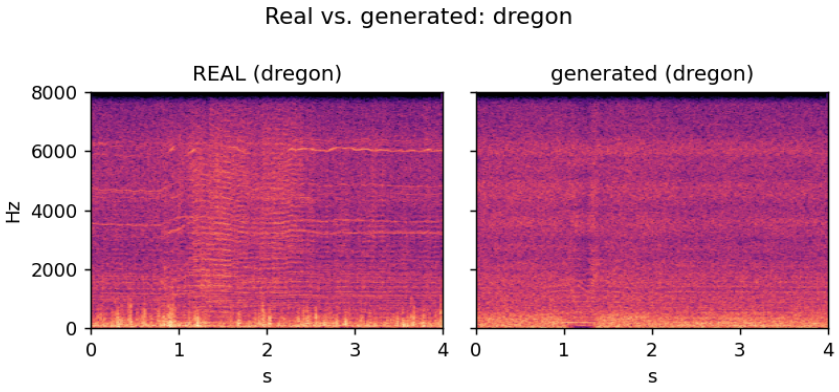
Cruise clip. All four track; the differences are precision: blind tracker 2.27, best neural 2.35, NMF 8.1 rev/s cruise MAE on the full split.

Why the frequency-scale augmentation is mandatory

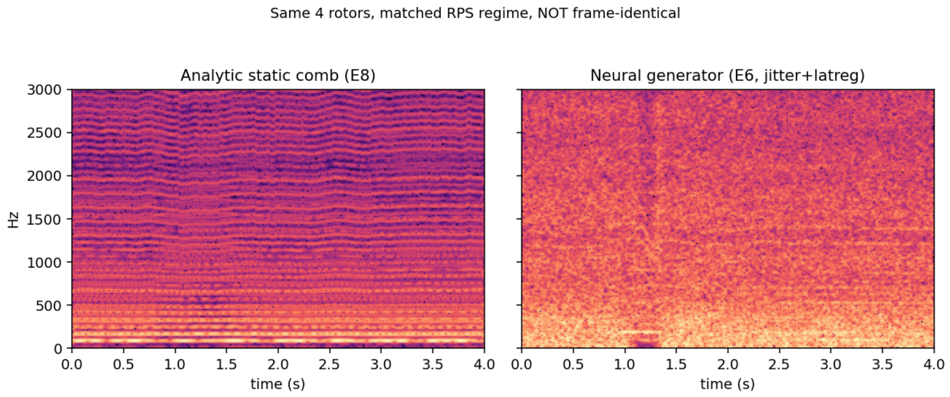


Models trained without it do not read frequency: a genuine 2% comb-spacing shift moves their prediction by 0.03–0.06% — they answer from loudness and the speed prior instead of the comb. The label-transforming frequency-scale augmentation destroys that shortcut and forces the model onto the harmonic structure.

Synthetic data, measured: generator vs comb



Neural generator output vs real DREGON noise: the timbre gap is audible and visible.



The analytic comb: guaranteed harmonic structure at exact labels — no generator training, no timbre modeling.

Pre-training on synthetic noise (R3), then fine-tuning on real data, was the best recipe of the July campaigns. The question the wrap-up answers: does it survive an equally rich **real** regime?

Synthetic data vs the full-envelope real regime

Run (SimpleConvV2 trunk)	zero	ramps	cruise	all
R2 real-only	**3.4**	**4.2**	**2.35**	**2.72**
R3 generated+comb curriculum	3.9	4.3	2.49	2.90
R5 mixed one-stage	16.1	9.1	5.20	7.11

Aggregate (val MSE)	SimpleConvV2	Transformer	causal GRU
R2 real-only control	**22.1**	**41.8**	61.9
R3 curriculum on R2	22.6	—	41.8
R4 comb-only on R2	—	—	**37.6**
R5 mixed on R2	147.6	59.2	85.8

For the best architecture the curriculum adds **nothing** (22.6 vs 22.1); mixed one-stage training loses on **all three** trunks. Synthetic data survives only for the weakest (causal) trunk, and there the free analytic comb beats the trained neural generator.

The verdict on generated data

- The July gains of the generated-first curriculum came from **coverage**: honest zeros and full-envelope trajectories the real corpus lacked — not from acoustic realism.
- The full-envelope real regime supplies the same coverage from real recordings alone (silence arm + SNR floor + label-transforming augmentations) — and the curriculum's advantage disappears.
- Mixing synthetic data into the real pool acts as label noise and degrades every architecture; staging is mandatory wherever synthetic data is used at all.
- Where a curriculum still pays (causal GRU), the analytic comb — zero training cost — beats the neural generator.